Flow Control and Error Control Numerical

Q1.
Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps and the propagation delay between the two nodes is 100 ms. Assume that the processing times at the source and destination are negligible and the ACK size is negligible. What is the channel utilization?

Q2.
A sliding-window protocol operates over a lossless link. Each frame is 1000 bits, bandwidth is 100 kbps, and one-way propagation delay is 100 ms. Processing time is zero and ACK transmission time is zero. What is the minimum window size required to achieve 100% link utilization?

Q3.
A sliding-window flow-control protocol operates over a full-duplex error-free link.

- Data frame = 2000 bits
- ACK = 10 bits
- Data rate = 1 Mbps
- One-way propagation delay = 100 ms
- Processing delays are negligible.

Find the minimum sender window size required to achieve 50% link utilization.

Q4.
A link has:
• Bandwidth = 1 Mbps
• Frame size = 1000 bits
• One-way propagation delay = 20 ms
• ACK transmission time = negligible

Find:
(a) Transmission time
(b) Minimum window size for 100% utilization
(c) Utilization when W = 10

Q5.

Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgement and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% is _____.

Q6.

Consider a selective repeat sliding window protocol that uses a frame size of 1KB to send data on 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is _____.

Q7.

Consider a 128 X 10^3 bits/second satellite communication link with one-way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____.